

Real-World Spectra Test Across All Trained Model Families

CNNIRPLASTICS — FTIR Polymer Classification

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Abstract

This report documents the work performed on 12 June 2026, in chronological order: (1) two cross-dataset generalization studies carried out before the branch merge — a blind evaluation of the saved CNN ensemble on the public FLOPP-e and BLoP reference libraries, and an A/B test of whether including the OpenSpecy library helps or hurts external generalization; (2) integration of the `add-cosine-baseline-eval` and the new `experiments` branches into `external-dataset-eval`, bringing the full staged-experiment pipeline and all trained model artifacts into one branch; and (3) a blind test of 30 newly measured real-world FTIR spectra against *every* trained model family in the repository, including per-sample saliency maps (gradient saliency and 1-D Grad-CAM) for the deployment CNN ensemble. The early studies motivated the six-class reject gate and the bioplastic-class extension that the later test exercises. The deployment configuration (random-forest soft-vote with a 0.89 reject gate) accepted only 4 of the 30 samples — `ldpewet`→HDPE, `pvcscrape`→PVC, `wetcd`→PS — and rejected every clear non-plastic, while the uncalibrated CNN softmax assigned 100% confidence to several out-of-distribution materials. The gate also rejected several plausible plastics (wet PP, LDPE-like films) on which all four modern model families agreed, suggesting the 0.89 threshold is conservative for spectra from this instrument.

1 Cross-dataset generalization (FLOPP-e and BLoP)

Before any retraining or merging, the saved CNN ensemble that was current at the start of the session (`output/fold_1..4.keras`, the four *pre-OpenSpecy* folds) was evaluated on two public FTIR reference libraries it had never seen, to measure transfer to other instruments and to materials outside the six trained classes:

- **FLOPP-e** (figshare 24593022): 59 reference polymer spectra, %T, $\sim 650\text{--}4000\text{ cm}^{-1}$ at 1 cm^{-1} .
- **BLoP**: 23 bioplastic spectra (PLA, PHA, CPLA, Bamboo, Sugarcane, Mater-Bi, Cornstarch, Flaxstic, Nature Flex), %T.

Each spectrum was interpolated onto the 1868-point c8 grid, per-spectrum min-max normalised on %T, and soft-voted across the four folds (`scripts/eval_external.py`). FLOPP-e “PE” pellets are counted correct as HDPE *or* LDPE, since the source does not split them; BLoP’s exact-zero low-wavenumber no-data region is dropped before interpolation.

In-distribution result. On the six commodity classes the ensemble reached **96.4 % (27/28 gradeable spectra: PE 11/11, PET 3/3, PP 7/7, PVC 2/2, PS 4/5)** on a completely different instrument — strong transfer for the clean reference library, in contrast to the messy real lab spectra tested later (Section 6).

Out-of-distribution behaviour. Of the 54 spectra outside the six classes, **27 were forced**

into a class at $\geq 90\%$ confidence: the bare softmax gives no novelty signal (PMMA $\rightarrow$ PS at 100 %, ABS $\rightarrow$ PS/PVC, etc.), and the BLoP bioplastics scattered across PS/PET/PVC. These two findings directly motivated (a) the explicit **reject gate** adopted in the deployment configuration and (b) the **bioplastic-class extension**. Artifacts: `results/external/`.

2 OpenSpecy inclusion: an A/B on generalization

A second pre-merge study tested the team’s hypothesis that the OpenSpecy library’s coarse 6 cm^{-1} sampling and its narrowed valid range cost more than its extra spectra give. Two Random-Forest arms were built with the project’s own loader, restricted to the six commodity classes, with identical model and identical per-spectrum min-max — the *only* deliberate difference being whether OpenSpecy is included (`scripts/ab_openspecy.py`).

Metric	without OpenSpecy	with OpenSpecy
Training spectra	6 000	14 124
Grid points	1 868	395
Grid range (cm^{-1})	399–4000	804–3168
CV accuracy (held-out folds)	99.85 %	97.26 %
External in-dist (FLOPP-e)	96.4 % (27/28)	100 % (28/28)
OOD forced @ $\geq 90\%$ conf	0/54	0/54

Table 1: OpenSpecy A/B (Random Forest, six commodity classes).

The headline finding is the **grid collapse**: forcing OpenSpecy in drops the shared grid from 1868 points ($399\text{--}4000\text{ cm}^{-1}$) to 395 points ($804\text{--}3168\text{ cm}^{-1}$) — a $\sim 79\%$ reduction and the loss of the entire $399\text{--}804\text{ cm}^{-1}$ low-wavenumber fingerprint region. The $+3.6\text{ pp}$ external gain is a single spectrum out of 28 (within noise), while in-domain CV *drops*, as expected for harder mixed data. The practical consequence is that the `experiments` branch keeps *both* configurations as separate model families — the no-OpenSpecy 1868-point `cnn-d1` and the with-OpenSpecy 395-point `cnn-os` (table 2) — so the trade-off can be judged on real data rather than assumed. Separately, the RF ensemble never crossed the 90 % bar on any OOD spectrum, foreshadowing the calibrated reject gate used below. Artifacts: `results/ab_openspecy/`.

3 Branch integration

After the generalization studies, the session continued by synchronising the local repository with `origin`. Two lines contained new work relative to `external-dataset-eval`:

- `add-cosine-baseline-eval` (7 commits): SLOPP and SLOPP-E reference datasets as CSV, a FLOPP dataset builder, and the class-set cleanup commits.
- `experiments` (2 commits, new branch): the staged experiment package `experiments/` (stage 1 data summary, stage 2 RF preprocessing sweep, stage 3 CNN/RF head-to-head, stage 4 stress/OOD testing), the full `experiments_output/` artifact tree with all trained models, cached aligned matrices, results JSON, figures, and `FINAL_REPORT`.

Both were merged into `external-dataset-eval` without conflicts (merge commits `0f1224f` and `77b852b`); `origin/main` and `origin/restructure` contained nothing new. The branch was pushed back to GitHub at the end of the session (`428a2dc`).

4 Model inventory

Every saved, loadable model family in the repository was tested (table 2). The two *legacy* families predate the experiments pipeline; the four modern families come from the merged **experiments** branch. The *deployment* configuration, as specified in `experiments_output/models/final/deployment.json`, is the random-forest 5-fold soft-vote with a reject rule: if the maximum class probability is below 0.89 the sample is declared UNKNOWN.

Family	Location	Models	Input grid	Preprocessing
legacy-cnn	<code>output/</code>	4 CNN folds	1868 pts	smooth+asls+snv
legacy-rf	<code>output/</code>	5 RF folds	600 pts	smooth+asls+snv
final-rf	<code>models/final/</code>	5 RF folds (+ <code>rf_final</code>)	1868 pts	snv, reject@0.89
final-cnn	<code>models/final/</code>	5 CNN folds	1868 pts	snv
cnn-d1	<code>models/no_os__smooth+d1+snv/</code>	5 CNN folds	1868 pts	smooth+d1+snv
cnn-os	<code>models/with_os__norm-snv/</code>	5 CNN folds	395 pts (804–3168 cm^{-1})	snv

Table 2: All tested model families. “Location” paths under `experiments_output/` where applicable. All families output the six commodity classes HDPE, LDPE, PP, PS, PVC, PET.

The Keras models were trained on Google Colab with a newer Keras version; a compatibility shim (stripping the unknown `quantization_config` layer kwarg, as already used in `scripts/saliency_cnn.py`) is required to load them locally and is included in the test script.

5 Test data and method

Thirty spectra measured on the lab FTIR were provided as two-column JCAMP-style text files (wavenumber in cm^{-1} , intensity in %T; 2,521 points from ~ 400 to $\sim 4000 \text{ cm}^{-1}$). The set deliberately mixes plastics, suspected bioplastics (black/grey printed “pla” samples), wet samples, and clear non-plastics (aluminium, paper, textile, rubber bands, a snack wrapper, the instrument background scan, thermoplastic polyurethane).

The pipeline implemented in `scripts/test_lab_downloads.py`:

1. parse each file, skipping `##` header lines;
2. align each spectrum to each family’s wavenumber grid by linear interpolation with edge clamping (the convention used for all external evaluations in the repo);
3. apply that family’s exact training preprocessing (per-spectrum, leakage-free: SNV, or smooth+AsLS+SNV, or smooth+1st-derivative+SNV);
4. soft-vote the fold ensemble of each family;
5. apply the deployment reject rule to the final-rf probabilities;
6. compute per-sample explainability maps for the deployment CNN ensemble: vanilla gradient saliency $|\partial p_{\hat{c}}/\partial x|$ and 1-D Grad-CAM on the last convolutional layer, averaged over the five folds, for the predicted class $\hat{c}$.

6 Results

Table 3 gives every prediction. The deployment model accepted **4 of 30** samples and declared the remaining 26 UNKNOWN.

6.1 Accepted samples

- **ldpewet** → **HDPE** (RF 96%, CNN 100%). A wet LDPE sample landing on HDPE is the familiar polyethylene-pair confusion; the density-related band differences (1377 cm^{-1} branch methyl, 730/719 split) are subtle and easily masked by water.
- **pvcscrape** → **PVC** (RF 92%; all four modern families 100%). The saliency map (figure 1) shows the ensemble keying on the $\sim 615\text{ cm}^{-1}$ C–Cl stretch region and the $1250\text{--}1430\text{ cm}^{-1}$ CH_2 deformation bands — chemically the right evidence.
- **wetcd** → **PS** (RF 89%, CNN 100%). Correct if the measured surface was the jewel case (polystyrene); the disc itself is polycarbonate, which is outside the six-class scope.

6.2 Out-of-distribution behaviour

Every clear non-plastic was rejected: aluminium, the background scan, paper, textiles, rubber bands, the snack wrapper, and all three TPU measurements. This matters because the CNN softmax alone was *confidently wrong* on several of them — **background** → PVC at 100%, **oreo** → PVC at 100%, **whitetextile** → HDPE at 100% — reproducing on fresh data the calibration warning recorded in **deployment.json** (CNN reject AUROC 0.836, softmax not usable as confidence). The TPU spectra produced maximally diffuse predictions across all families (35–45%), textbook OOD behaviour. The black and grey printed samples (**blackpla***, **greypla***, **wetpla**) drew low-confidence, mutually contradictory predictions from every family (PVC $\sim 55\%$ / PS 60% / HDPE 40–90%); if these are PLA prints the rejection is correct, and carbon-black-filled samples also genuinely yield washed-out ATR spectra.

6.3 Borderline rejections

The gate also rejected samples that look like genuine commodity plastics on which all four modern families agree:

- **wetpp** → PP (CNN 96%, d1-CNN 100%, OS-CNN 87%) but RF only 69%;
- **pop/pop2** → LDPE (three CNN families 97–100%) but RF 76–80%;
- **yellowpapturnd** → PVC at 98–100% on three families (likely the coating/laminate side of the paper), RF 58%.

Wet or contaminated surfaces depress the RF vote below 0.89, so the deployment threshold — tuned on FLOPP-e/BLoP — trades these away. If ground truth can be confirmed for these files, this set is a natural calibration set for re-tuning the threshold to this instrument.

6.4 Legacy models

The legacy families are not usable on this data. The legacy CNN collapses to PVC at $\sim 100\%$ for most inputs regardless of content, and the legacy RF never exceeds 37% confidence (it expects a different 600-point grid and an older training distribution). The retraining done in the **experiments** branch was clearly worthwhile.

7 Saliency maps

For each sample, gradient saliency and 1-D Grad-CAM were computed for the deployment CNN ensemble’s predicted class and overlaid on the preprocessed spectrum (folder **saliency/** next to this report). Figure 4 stacks all 30 Grad-CAM profiles.

File	Deployment	legacy-cnn	legacy-rf	final-rf	final-cnn	cnn-d1	cnn-os
allumium	UNKNOWN	PVC 100%	PP 31%	PVC 60%	PVC 47%	PVC 100%	PET 73%
background	UNKNOWN	HDPE 100%	HDPE 26%	PVC 54%	PVC 100%	PVC 100%	PET 83%
blaclpla	UNKNOWN	PVC 75%	PP 30%	PET 37%	PS 50%	HDPE 55%	PS 60%
blaclpla2	UNKNOWN	PVC 75%	PP 31%	PET 37%	PVC 55%	HDPE 72%	PS 60%
blaclpla3	UNKNOWN	PVC 75%	PP 30%	PET 37%	PVC 57%	HDPE 39%	PS 60%
brown	UNKNOWN	PVC 78%	PP 24%	PVC 67%	HDPE 100%	HDPE 79%	HDPE 70%
brown2	UNKNOWN	PVC 82%	PP 24%	PVC 67%	HDPE 99%	HDPE 82%	HDPE 51%
greypla	UNKNOWN	PVC 75%	PP 30%	PET 37%	PVC 52%	HDPE 76%	PS 60%
greypla2	UNKNOWN	PVC 75%	PP 32%	PET 37%	PVC 60%	PET 41%	PS 60%
greypla3	UNKNOWN	PVC 75%	PP 31%	PET 37%	PVC 59%	HDPE 74%	PS 59%
greypla4	UNKNOWN	PVC 75%	PP 30%	PET 37%	PVC 59%	HDPE 45%	PS 59%
ldpewet	HDPE	PET 41%	PS 25%	HDPE 96%	HDPE 100%	HDPE 67%	HDPE 79%
orangerubberband	UNKNOWN	PVC 100%	PP 26%	PP 77%	PP 100%	PP 100%	PP 98%
orangerubberband2	UNKNOWN	PVC 100%	PP 33%	PP 77%	PP 100%	PP 76%	PP 99%
oreo	UNKNOWN	PVC 100%	PP 37%	PVC 64%	PVC 100%	PVC 100%	PP 43%
oreo2	UNKNOWN	PVC 100%	PP 37%	PVC 62%	PVC 88%	PVC 100%	PVC 50%
paper1	UNKNOWN	PVC 74%	PET 33%	PVC 73%	HDPE 78%	PVC 58%	HDPE 96%
pop	UNKNOWN	PET 50%	HDPE 28%	LDPE 76%	LDPE 100%	LDPE 100%	LDPE 98%
pop2	UNKNOWN	PET 53%	HDPE 28%	LDPE 80%	LDPE 100%	LDPE 100%	LDPE 97%
pvcscrape	PVC	PVC 100%	PP 37%	PVC 92%	PVC 100%	PVC 100%	PVC 100%
TPU	UNKNOWN	PVC 100%	HDPE 27%	PET 45%	PET 37%	PVC 61%	PET 96%
TPU2	UNKNOWN	PVC 100%	HDPE 26%	PET 45%	PVC 38%	PVC 62%	PET 95%
TPU3	UNKNOWN	PVC 100%	HDPE 27%	PET 44%	PS 35%	PVC 68%	PET 97%
wetcd	PS	PVC 54%	PP 31%	PS 89%	PS 100%	PS 100%	PS 100%
wetpla	UNKNOWN	PVC 75%	PP 31%	PET 32%	PVC 79%	HDPE 94%	PS 51%
wetpp	UNKNOWN	PVC 34%	HDPE 24%	PP 69%	PP 96%	PP 100%	PP 87%
whitetextile	UNKNOWN	HDPE 75%	PP 25%	PVC 61%	HDPE 100%	HDPE 59%	HDPE 69%
whitetextile2	UNKNOWN	HDPE 100%	PET 22%	PVC 61%	HDPE 100%	HDPE 68%	HDPE 56%
yellowpap	UNKNOWN	HDPE 100%	PP 25%	PP 57%	PP 81%	PP 100%	PP 96%
yellowpapturnd	UNKNOWN	PVC 100%	HDPE 27%	PVC 58%	PVC 98%	PVC 100%	PVC 99%

Table 3: Every model family on every spectrum: predicted class and ensemble confidence. **Deployment** = final-rf soft-vote with the 0.89 reject gate from `deployment.json`.

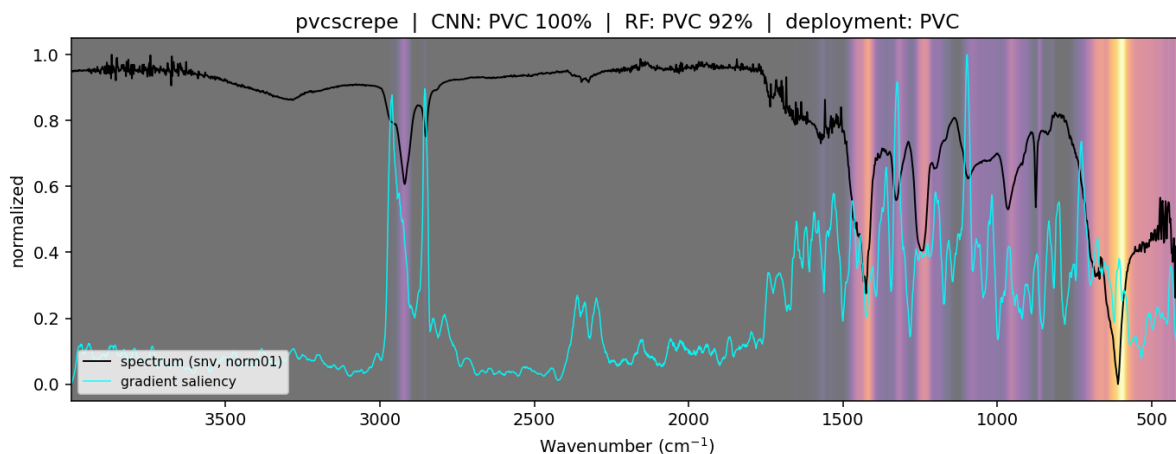


Figure 1: `pvcscrape`: attention concentrates on the C–Cl stretch region ($\sim 615\text{ cm}^{-1}$) and the $1250\text{--}1430\text{ cm}^{-1}$ bands — genuine PVC evidence.

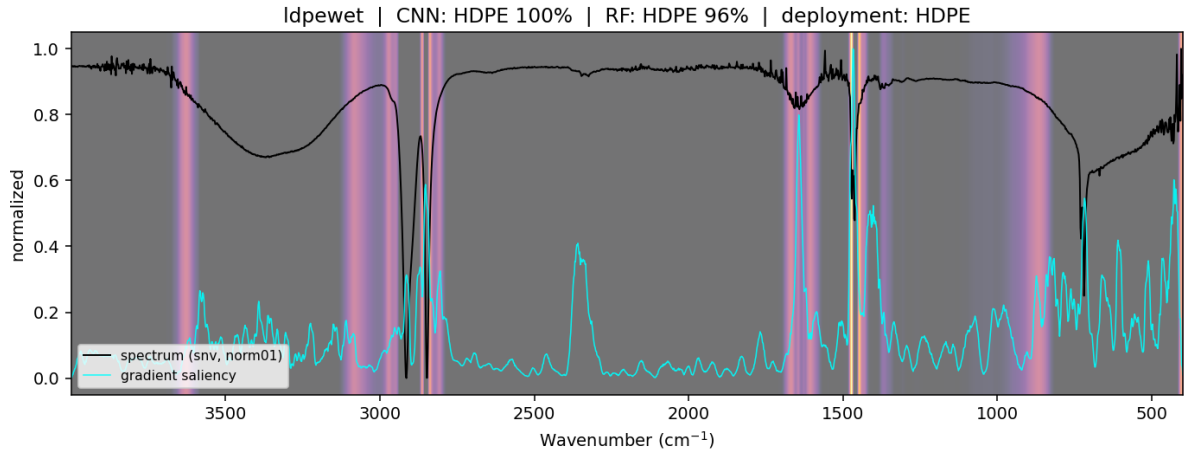


Figure 2: **ldpewet** (accepted as HDPE): weight on the CH_2 stretch/bend regions shared by both polyethylenes; the discriminating LDPE features are weak, consistent with the HDPE/LDPE swap.

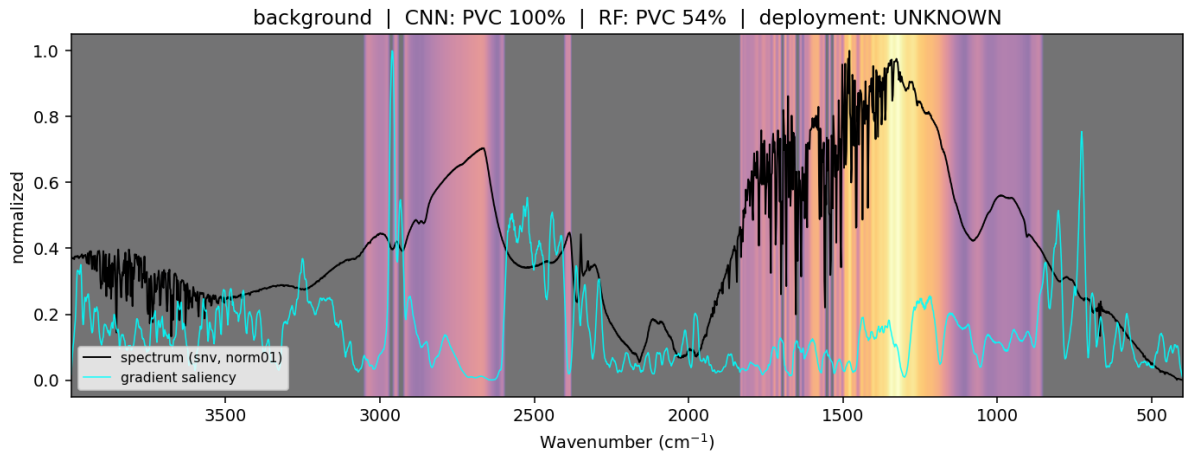
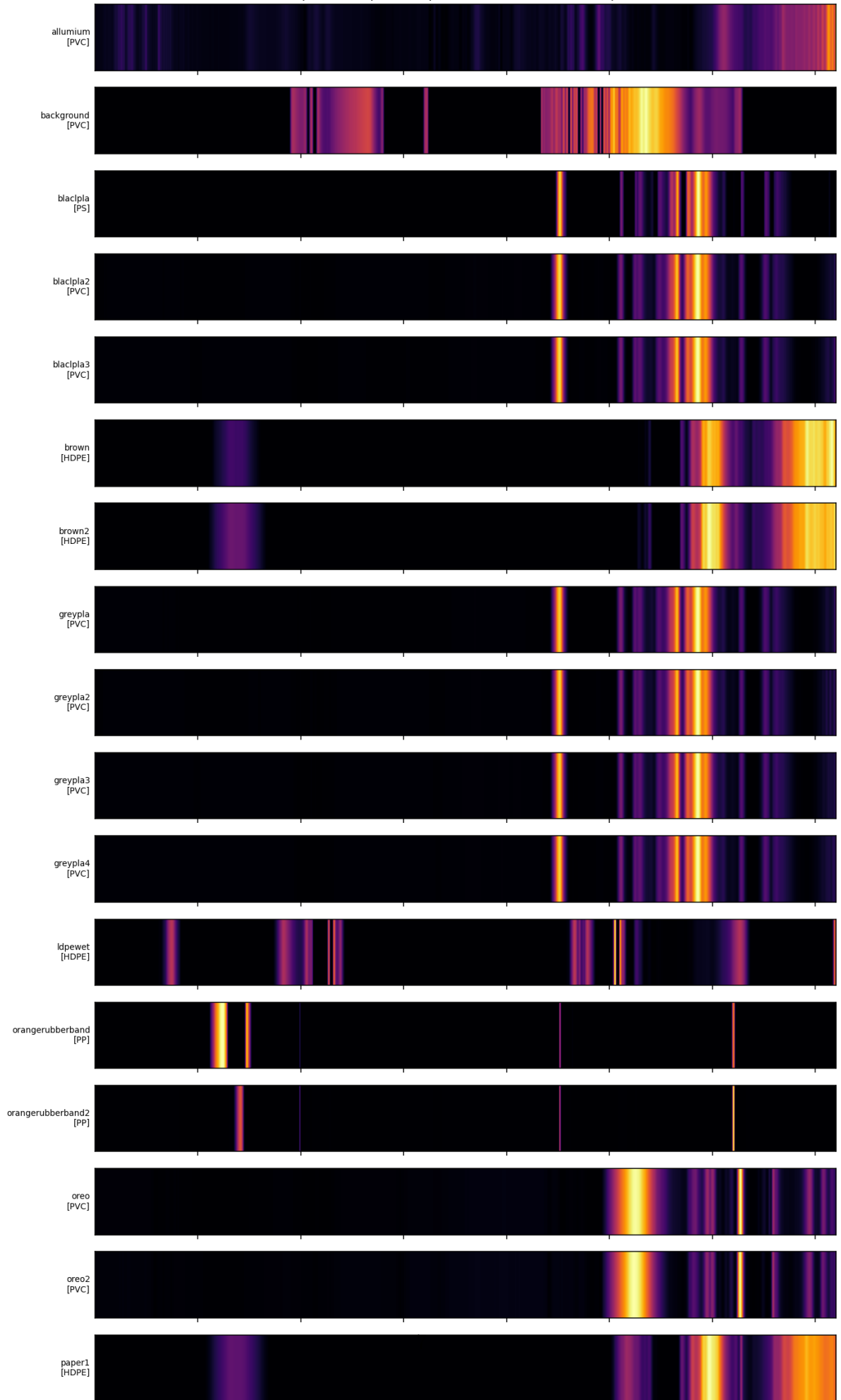


Figure 3: **background** (instrument background scan): the CNN reports PVC at 100% while attending to broad, featureless regions — a clear illustration of why the uncalibrated softmax must sit behind the RF reject gate.

Grad-CAM importance per sample (final CNN ensemble, predicted class)



8 Conclusions and next steps

1. The merged deployment pipeline behaves as designed on completely fresh real-world data: it never produced a confident wrong answer, at the cost of high abstention (26/30 UNKNOWN).
2. The CNN softmax over-confidence on OOD inputs is reproduced (100% on a background scan); the RF gate is what makes the system safe.
3. The 0.89 threshold appears too strict for wet/contaminated samples from this instrument; **wetpp**, **pop**, and **yellowpapturnd** are candidates for threshold re-calibration once their ground truth is confirmed.
4. The legacy **output/** models should be considered deprecated for evaluation purposes.
5. The earlier generalization studies underwrite the design tested here: the FLOPP-e/BLoP out-of-distribution over-confidence motivated the reject gate (whose 0.89 threshold was tuned on those same datasets), and the OpenSpecy grid-collapse A/B justifies carrying both the 1868-point **cnn-d1** and the 395-point **cnn-os** families forward.
6. All artifacts are committed on **external-dataset-eval** (428a2dc): the test script **scripts/test_lab_dow**, the cross-dataset and A/B scripts (**scripts/eval_external.py**, **scripts/ab_openspecy.py**), the prediction tables, and all 30 saliency maps.